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# Network Forensics and Protocols

Prof Bill Buchanan OBE FRSE

# Outline

- The Core Protocols.
- ARP.
- SYN.
- FTP.
- ICMP.
- DNS.
- Port Scans.
- SYN Floods.
- Application Layer.



O

- Th
- AF
- SY
- FT
- IC
- DI
- Pc
- SY
- Ap

## PCap File Analysis

[Network Forensics Home][Home]

This is a page for a PCAP file analysis. [Install Wireshark][Network Forensics Test] [Network Forensics Test (Mill)] [HTTP Analyser][SSL/TSL] Here are some examples:

Digital Forensics
FFD8 47 49 46 39 PK
@asecuritysite.com

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Web Page** | [Go View Video] | **Google Web** | [Go View Video] | **DNS Lookup** | [Go View Video] | **IPSec** | [Go View] |
| **FTP Access** | [Go View Video] | **FTP 2** | [Go View] | **Telnet** | [Go View] | **IP (Windows)** | [Go View] |
| **Ping** | [Go View] | **Tracert** | [Go View] | **Spoof Address** | [Go View Video] | **SSL** | [Go View] |
| **Trace Route** | [Go View] | **SMTP** | [Go View] | **IMAP** | [Go View] | **POP-3** | [Go View] |
| **NMAP** | [Go View Video] | **NMAP (FIN)** | [Go View Video] | **NMAP (XMAS) [3]** | [Go View] | **NMAP (UDP)** | [Go View] |
| **NMAP (Null)** | [Go View] | **ARP Sweep [1]** | [Go View] | **Ping Sweep [2]** | [Go View] | **Hydra FTP** | [Go View Video ] |
| **Hydra Telnet** | [Go View Video] | **Hping SYN** | [Go View Video] | **Hping FIN** | [Go View Video] | **Hping Scan** | [Go View Video] |
| **Hping UDP Scan** | [Go View Video] | **Hping Port 80** | [Go View] | **Hping FIN (80)** | [Go View] | **SMB** | [Go View] |
| **IP (Ubuntu)** | [Go View] | **Malware Trace** | [Go View Video] | **Web Page 2** | [Go View] | **ARP Spoof** | [Go View] |
| **My SQL** | [Go View] | **NTP** | [Go View] | **SNMP** | [Go View] | **RIP** | [Go View] |
| **TFTP** | [Go View] | **DHCP** | [Go View] | **Heartbleed** | [Go View] | **Syslog** | [Go View] |
| **Teardrop** | [Go View] | **Bitorrent** | [Go View] | **PNG** | [Go View] | **JPEG** | [Go View] |
| **ZIP** | [Go View] | **Credit card** | [Go View] | **Flash** | [Go View] | **GIF** | [Go View] |
| **Wanna Decryptor** | [Go View] | **Android (WebView)** | [Go View Det] | **Reverse TCP** | [Go View | **Android Malware** | [Go View |

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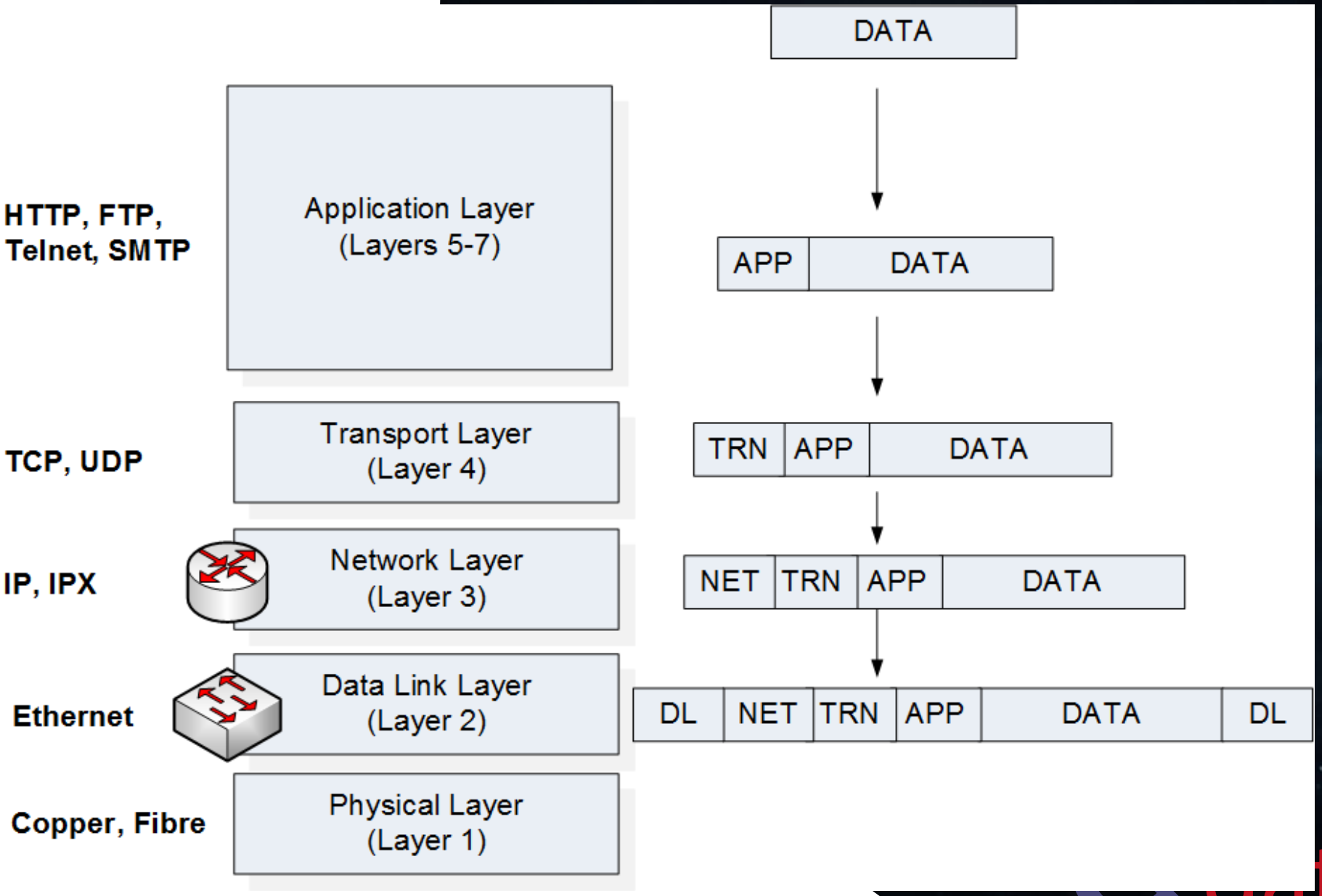
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# The Core Protocols

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# Data Encapsulation



# Ethernet, IP and TCP

**Protocol:**
**1 – ICMP**
**6 – TCP**
**8 – EGP**
**17 - UDP**

**IP header**

| Version | Header len. | Type of service |
|---|---|---|
| Total length | | |
| Identification | | |
| 0 | D | M | Fragment Offset |
| Time-to-live (TTL) | | Protocol |
| Header Checksum | | |
| Source IP Address | | |
| Destination IP Address | | |

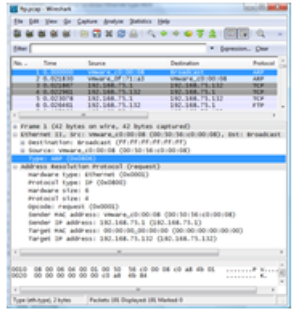
**TCP header**

| TCP Source Port | |
|---|---|
| TCP Destination Port | |
| Sequence Number | |
| Acknowledgement Number | |
| Data Offset | Flags/Reserved |
| Window | |
| Checksum | |
| Urgent Pointer | |

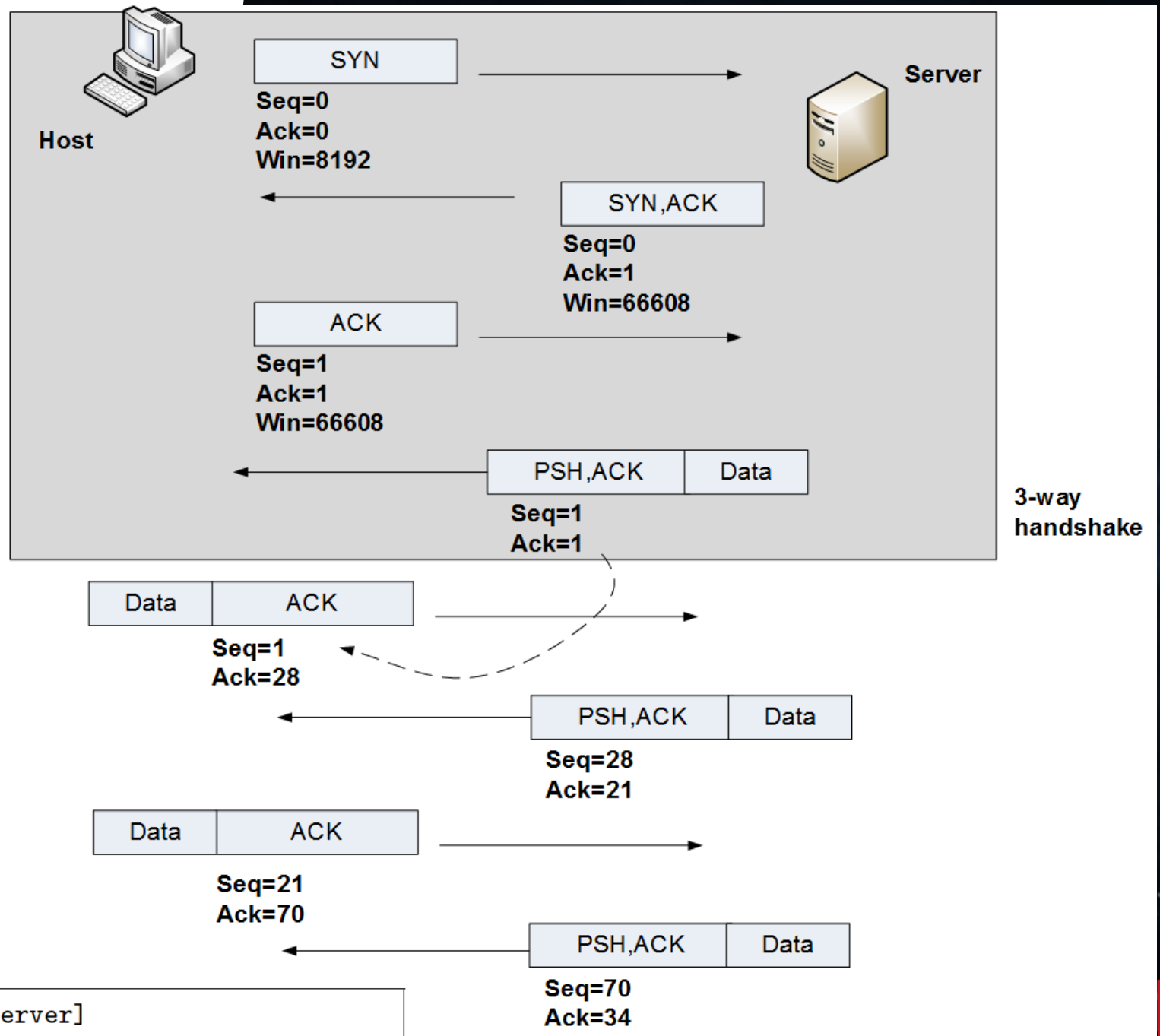
| Preamble (7 bytes) | Start (1 B) | Dest. MAC (6 bytes) | Src. MAC (6 bytes) | Type (2 bytes) | data... | Checksum (4 bytes) |
|---|---|---|---|---|---|---|

**Type:**
**0x800 – IP**
**0x806 – ARP**

**Ethernet frame**



# Three-way handshake




```
IP[Host]:TCPport[Host] -> IP[Server]:TCPport[Server]
```

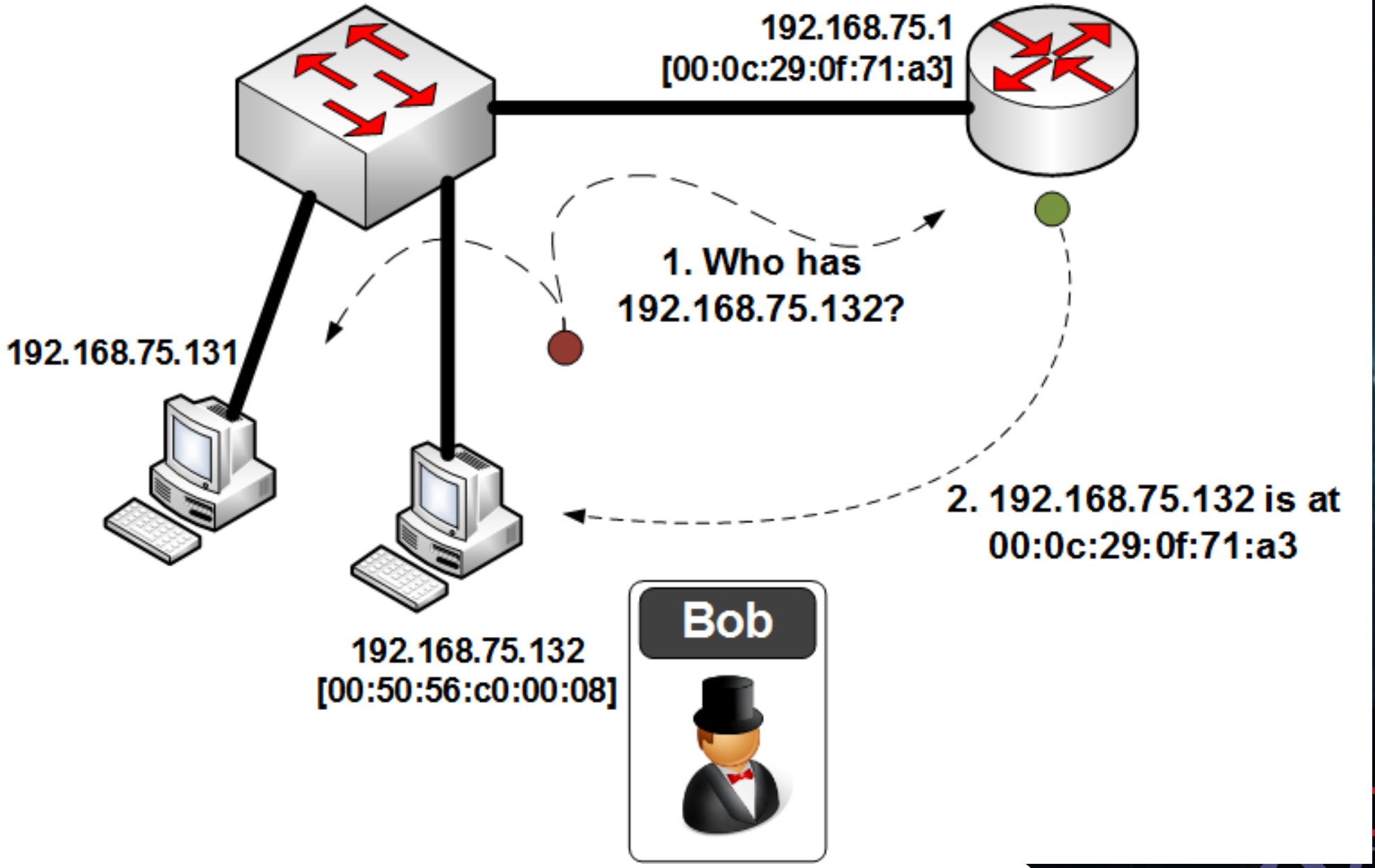
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# ARP

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# ARP

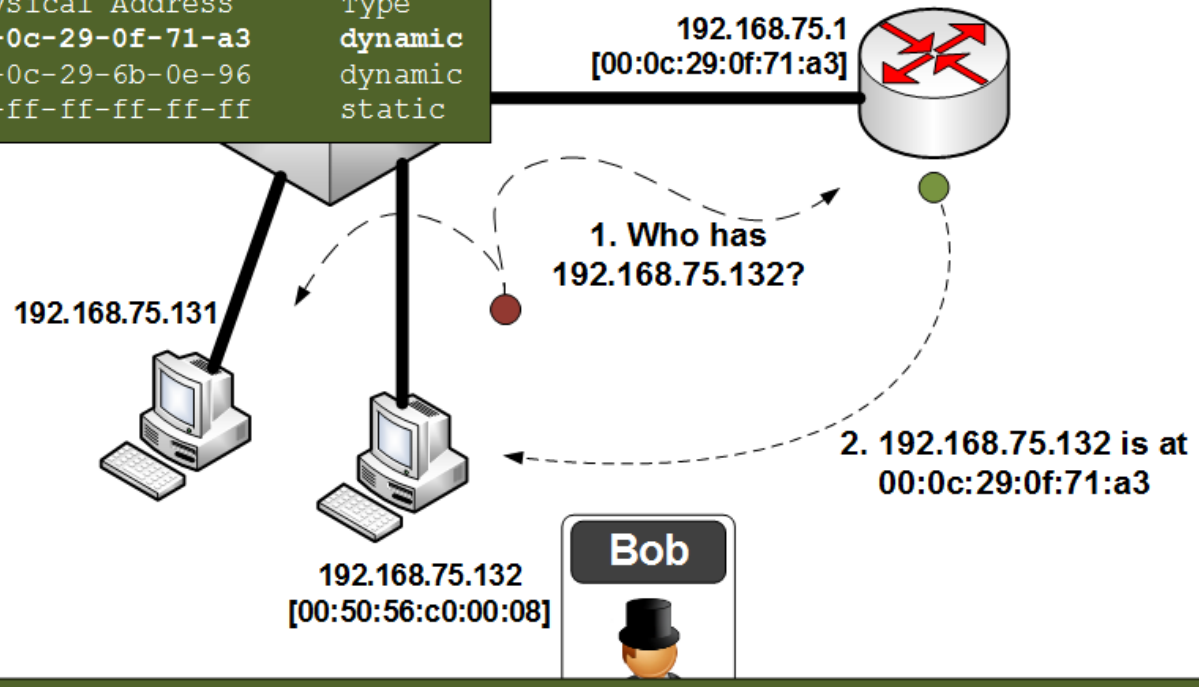

192.168.75.1
[00:0c:29:0f:71:a3]
1. Who has
192.168.75.132?
192.168.75.131
2. 192.168.75.132 is at
00:0c:29:0f:71:a3
192.168.75.132
[00:50:56:c0:00:08]
Bob

# ARP

```
No.     Time        Source                Destination           Protocol Info
      1 0.000000    Vmware_c0:00:08       Broadcast             ARP      Who has
192.168.75.132?  Tell 192.168.75.1

Frame 1 (42 bytes on wire, 42 bytes captured)
Ethernet II, Src: Vmware_c0:00:08 (00:50:56:c0:00:08), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Address Resolution Protocol (request)
```

```
Interface: 192.168.75.1 --- 0x1d
  Internet Address      Physical Address      Type
  192.168.75.132        00-0c-29-0f-71-a3     dynamic
  192.168.75.138        00-0c-29-6b-0e-96     dynamic
  192.168.75.255        ff-ff-ff-ff-ff-ff     static
```




```
No.     Time        Source                Destination           Protocol Info
      2 0.021830    Vmware_0f:71:a3       Vmware_c0:00:08       ARP      192.168.75.132 is at
00:0c:29:0f:71:a3

Frame 2 (42 bytes on wire, 42 bytes captured)
Ethernet II, Src: Vmware_0f:71:a3 (00:0c:29:0f:71:a3), Dst: Vmware_c0:00:08 (00:50:56:c0:00:08)
Address Resolution Protocol (reply)
```

ARP

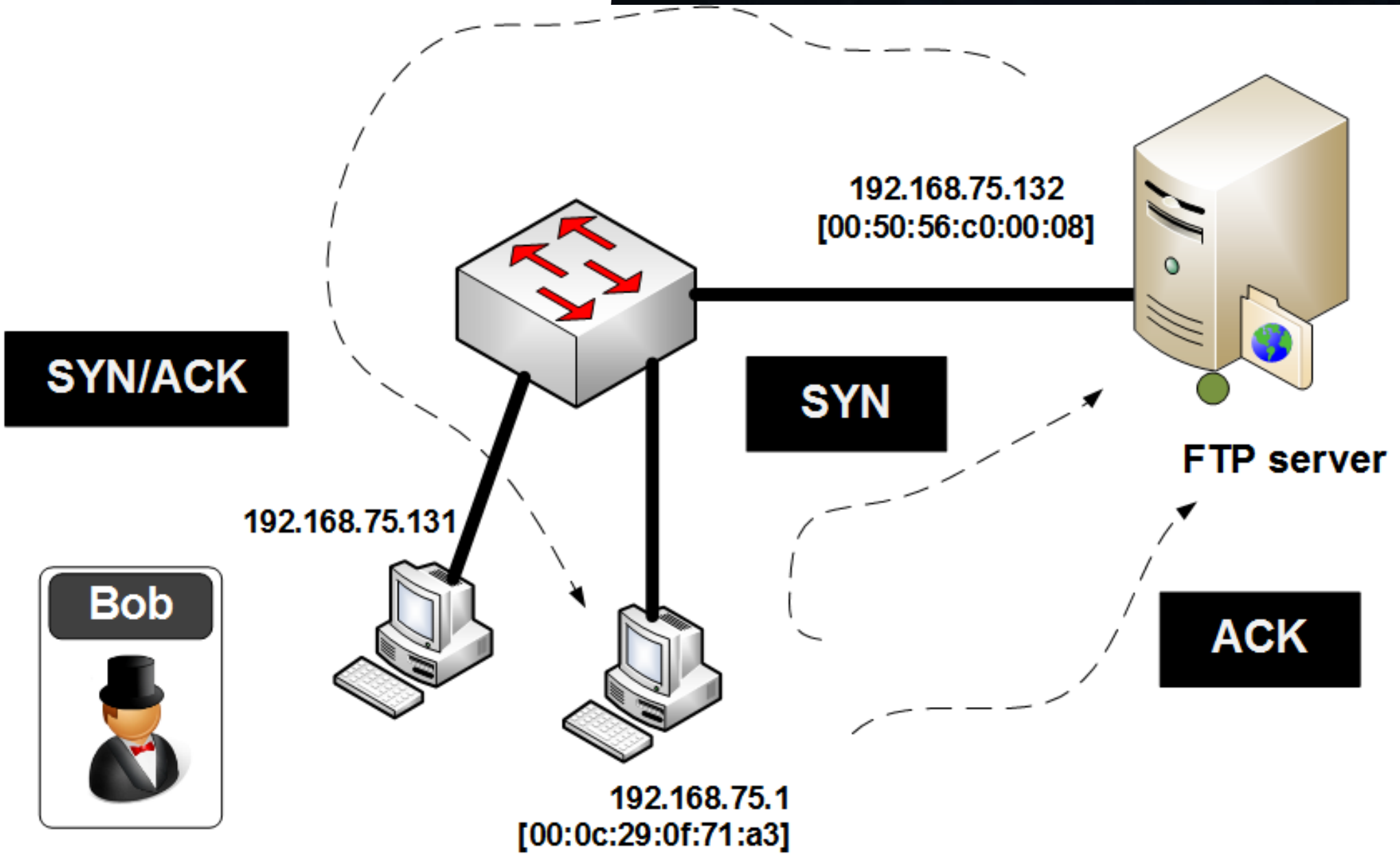
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# TCP - SYN

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# SYN


192.168.75.132
[00:50:56:c0:00:08]
SYN/ACK
SYN
FTP server
192.168.75.131
Bob
ACK
192.168.75.1
[00:0c:29:0f:71:a3]

SYN

# SYN

```
No.     Time        Source                Destination           Protocol Info
      3 0.021867    192.168.75.1          192.168.75.132        TCP      abatemgr > ftp [SYN] Seq=0
Win=8192 Len=0 MSS=1460 WS=2 TSV=683746 TSER=0

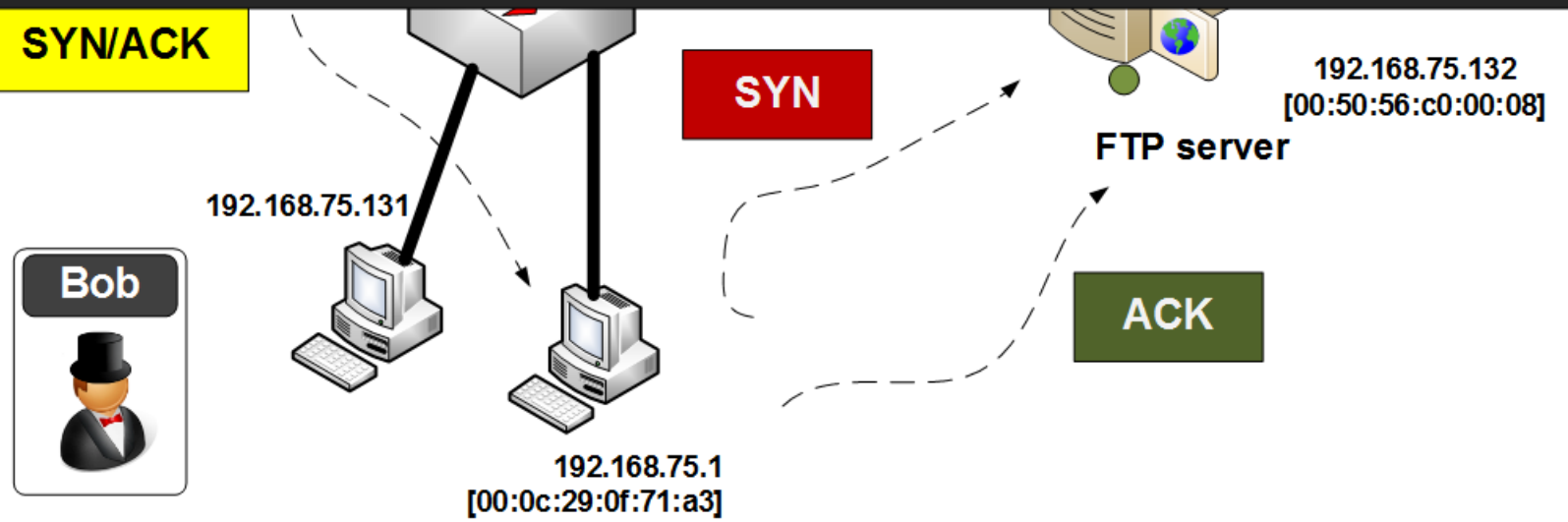
Frame 3 (74 bytes on wire, 74 bytes captured)
Internet Protocol, Src: 192.168.75.1 (192.168.75.1), Dst: 192.168.75.132 (192.168.75.132)
Transmission Control Protocol, Src Port: abatemgr (3655), Dst Port: ftp (21), Seq: 0, Len: 0

No.     Time        Source                Destination           Protocol Info
      4 0.022961    192.168.75.132        192.168.75.1          TCP      ftp > abatemgr [SYN, ACK]
Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=0 TSV=0 TSER=0

Frame 4 (78 bytes on wire, 78 bytes captured)
Internet Protocol, Src: 192.168.75.132 (192.168.75.132), Dst: 192.168.75.1 (192.168.75.1)
Transmission Control Protocol, Src Port: ftp (21), Dst Port: abatemgr (3655), Seq: 0, Ack: 1, Len: 0

No.     Time        Source                Destination           Protocol Info
      5 0.023078    192.168.75.1          192.168.75.132        TCP      abatemgr > ftp [ACK] Seq=1
Ack=1 Win=66608 Len=0 TSV=683748 TSER=0

Frame 5 (66 bytes on wire, 66 bytes captured)
Internet Protocol, Src: 192.168.75.1 (192.168.75.1), Dst: 192.168.75.132 (192.168.75.132)
Transmission Control Protocol, Src Port: abatemgr (3655), Dst Port: ftp (21), Seq: 1, Ack: 1, Len: 0
```




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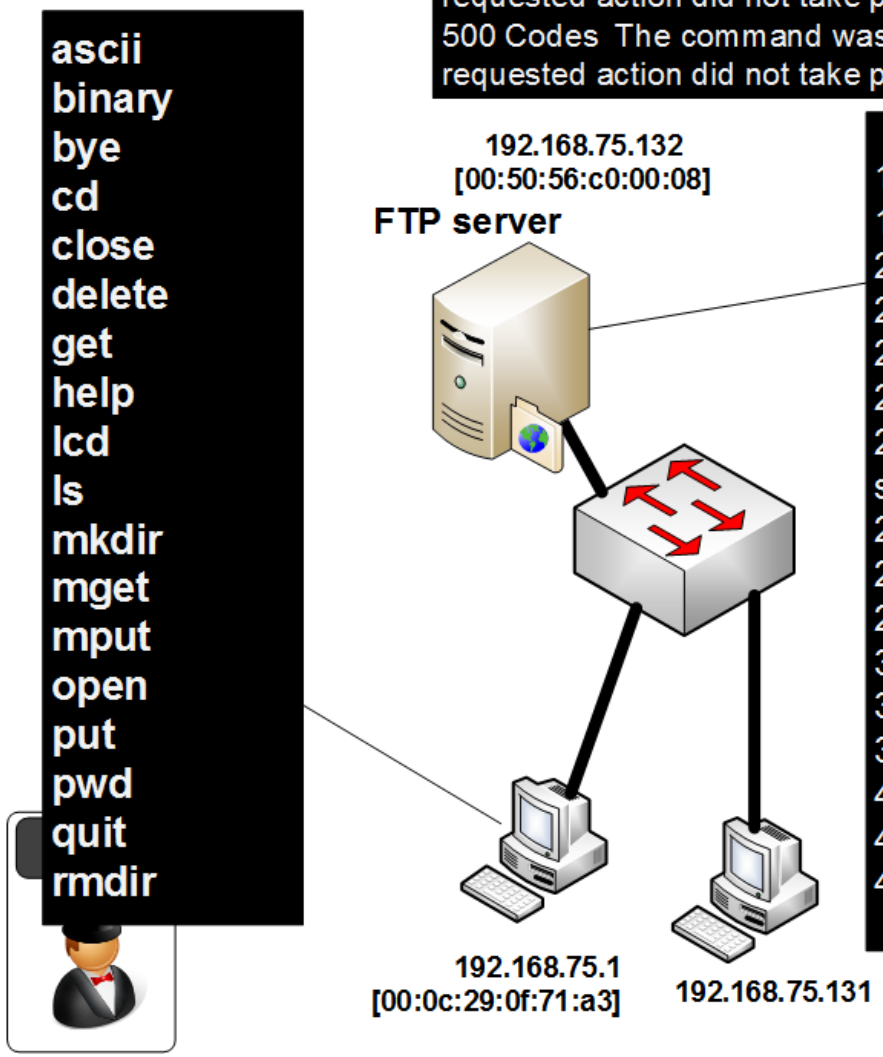
# FTP

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# FTP

100 Codes The requested action is being taken.
200 Codes The requested action has been successfully completed.
300 Codes The command has been accepted, but the requested action is being held pending receipt of further information.
400 Codes The command was not accepted and the requested action did not take place.
500 Codes The command was not accepted and the requested action did not take place.

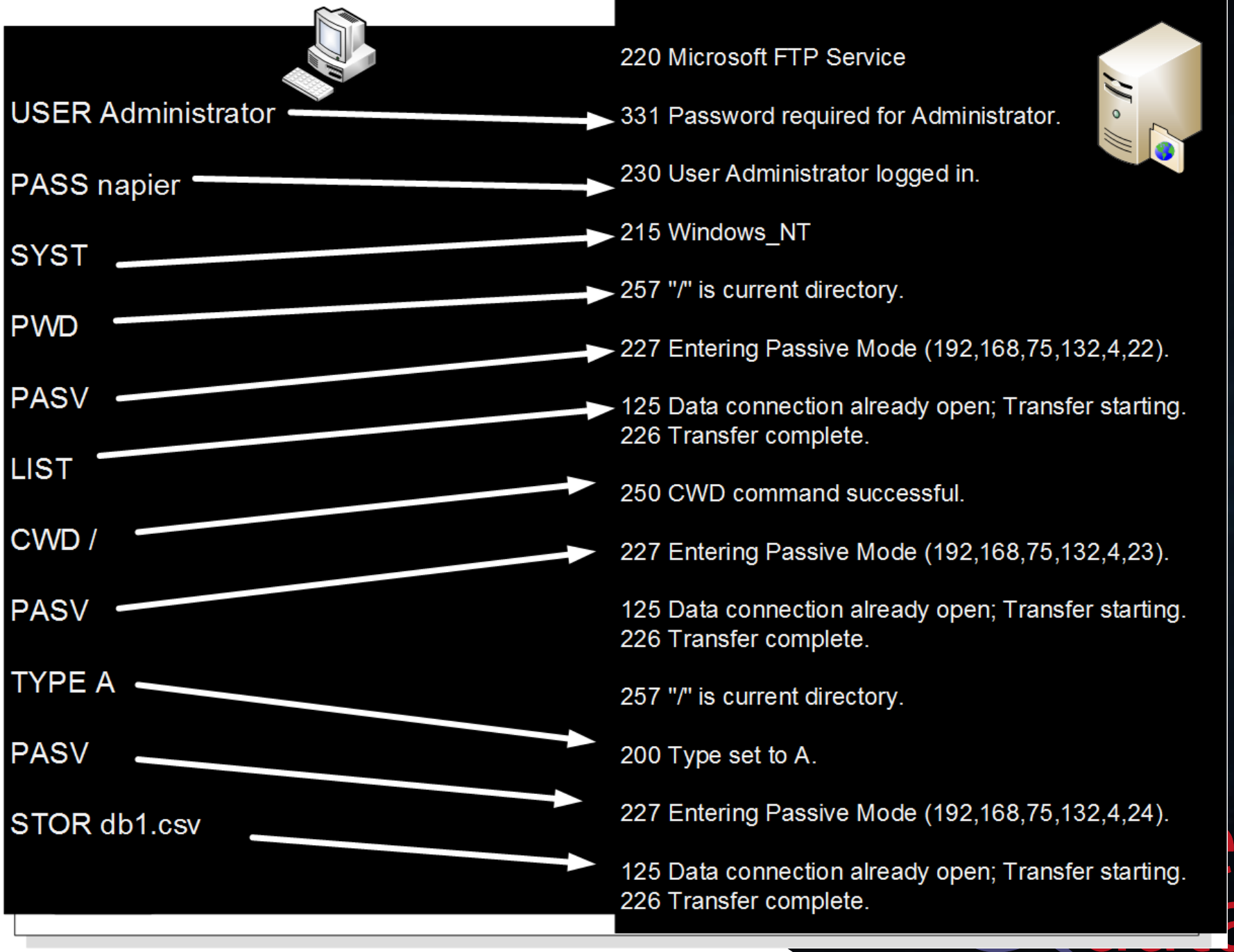
**ascii**
**binary**
**bye**
**cd**
**close**
**delete**
**get**
**help**
**lcd**
**ls**
**mkdir**
**mget**
**mput**
**open**
**put**
**pwd**
**quit**
**rmdir**



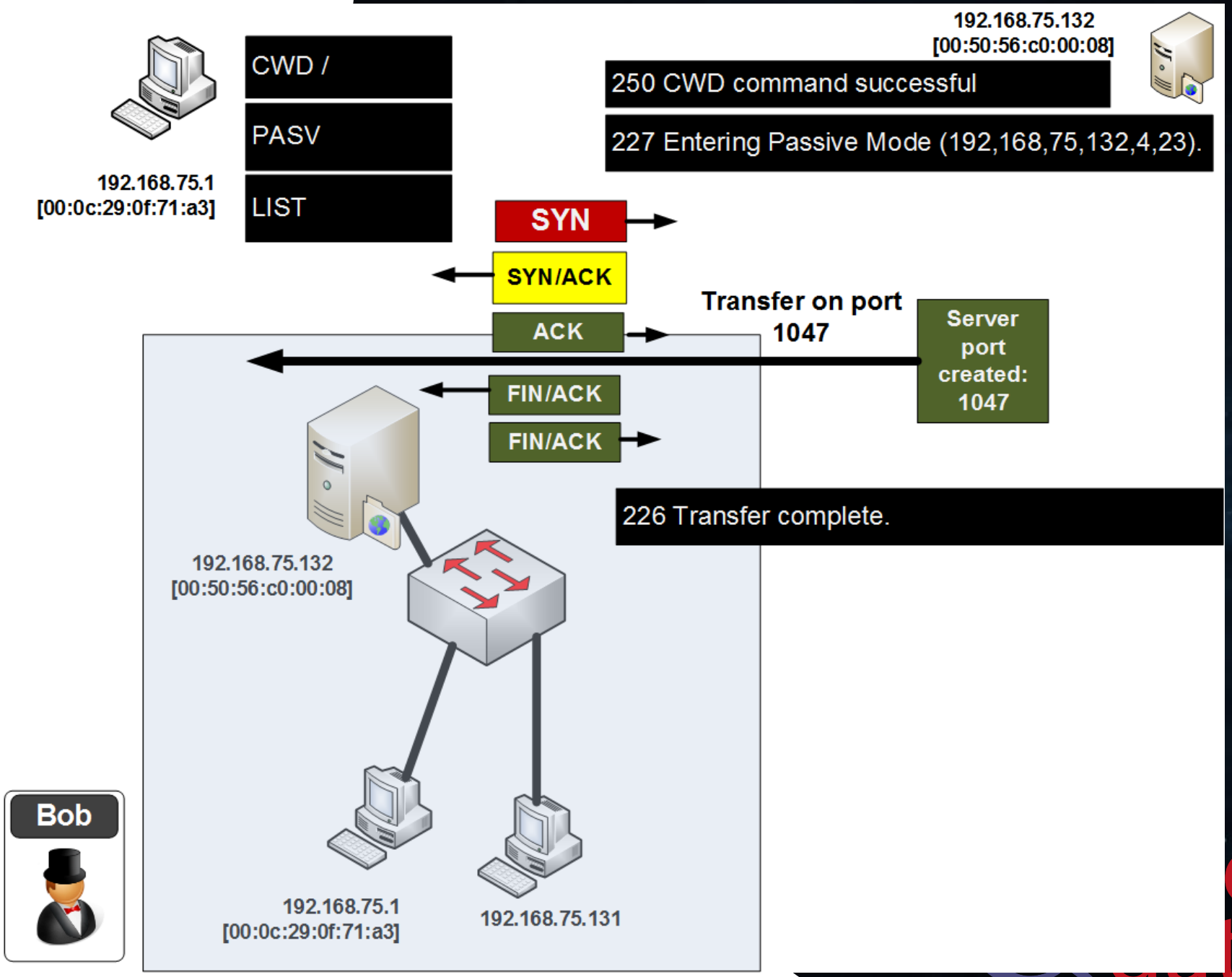

125 Data connection already open, transfer starting.
150 File status okay, about to open data connection.
200 Command okay.
202 Command not implemented
211 System status, or system help reply.
212 Directory status.
226 Closing data connection. Requested file action successful (file transfer, abort, etc.).
227 Entering Passive Mode
230 User logged in, proceed.
250 Requested file action okay, completed.
331 User name okay, need password.
332 Need account for login.
350 Requested file action pending further information.
421 Service not available, closing control connection.
425 Can't open data connection.
426 Connection closed, transfer aborted.

FTP

# FTP


220 Microsoft FTP Service
USER Administrator
331 Password required for Administrator.
PASS napier
230 User Administrator logged in.
SYST
215 Windows_NT
PWD
257 "/" is current directory.
PASV
227 Entering Passive Mode (192,168,75,132,4,22).
LIST
125 Data connection already open; Transfer starting.
226 Transfer complete.
CWD /
250 CWD command successful.
PASV
227 Entering Passive Mode (192,168,75,132,4,23).
TYPE A
125 Data connection already open; Transfer starting.
226 Transfer complete.
PASV
257 "/" is current directory.
STOR db1.csv
200 Type set to A.
227 Entering Passive Mode (192,168,75,132,4,24).
125 Data connection already open; Transfer starting.
226 Transfer complete.

# FTP


192.168.75.132
[00:50:56:c0:00:08]
CWD /
250 CWD command successful
PASV
227 Entering Passive Mode (192,168,75,132,4,23).
LIST
192.168.75.1
[00:0c:29:0f:71:a3]
SYN
SYN/ACK
ACK
Transfer on port
1047
Server
port
created:
1047
FIN/ACK
FIN/ACK
226 Transfer complete.
192.168.75.132
[00:50:56:c0:00:08]
Bob
192.168.75.1
[00:0c:29:0f:71:a3]
192.168.75.131

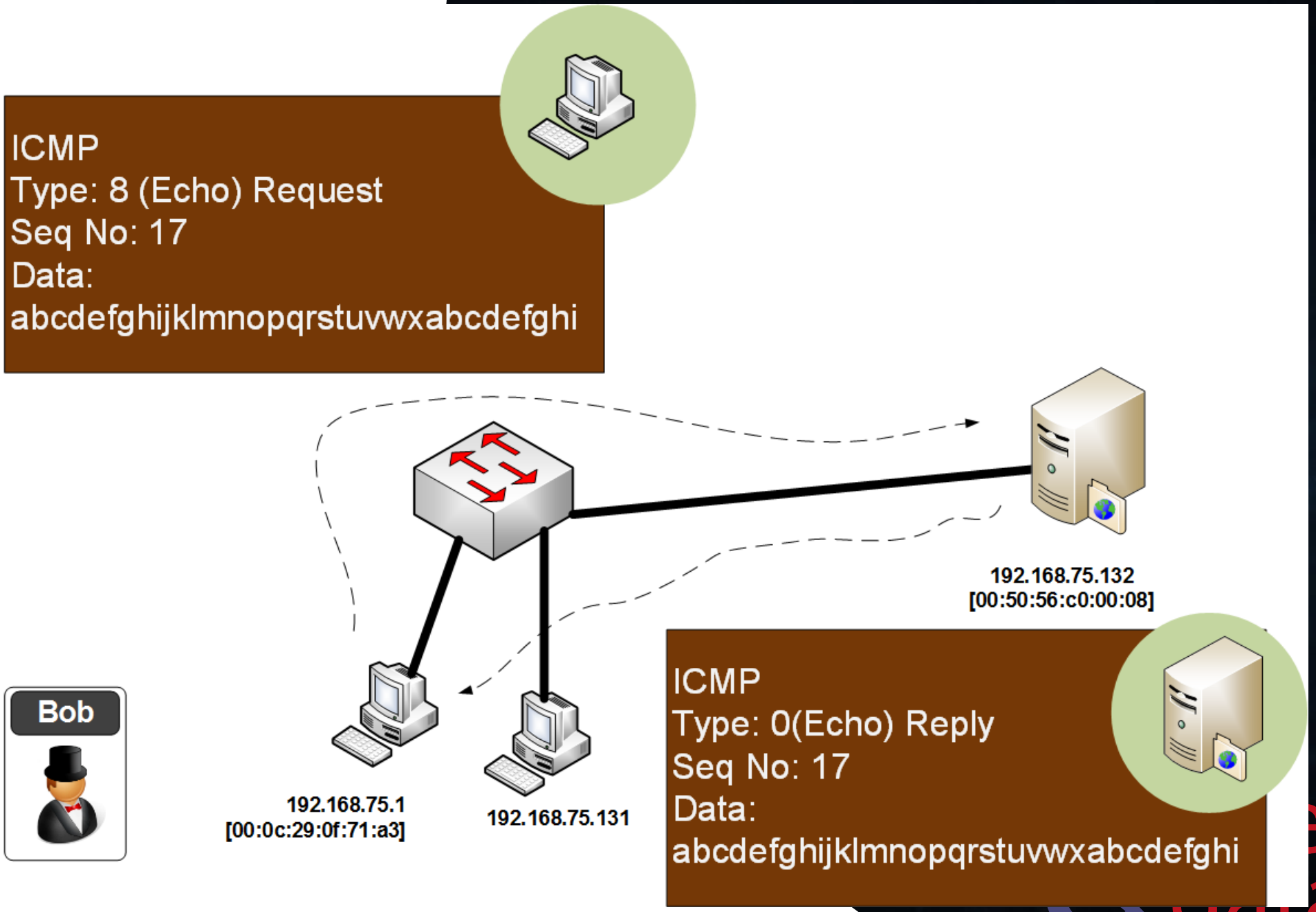
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# ICMP

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# ICMP


ICMP
Type: 8 (Echo) Request
Seq No: 17
Data:
abcdefghijklmnopqrstuvwxabcdefghi
192.168.75.132
[00:50:56:c0:00:08]
Bob
192.168.75.1
[00:0c:29:0f:71:a3]
192.168.75.131
ICMP
Type: 0(Echo) Reply
Seq No: 17
Data:
abcdefghijklmnopqrstuvwxabcdefghi

ICMP

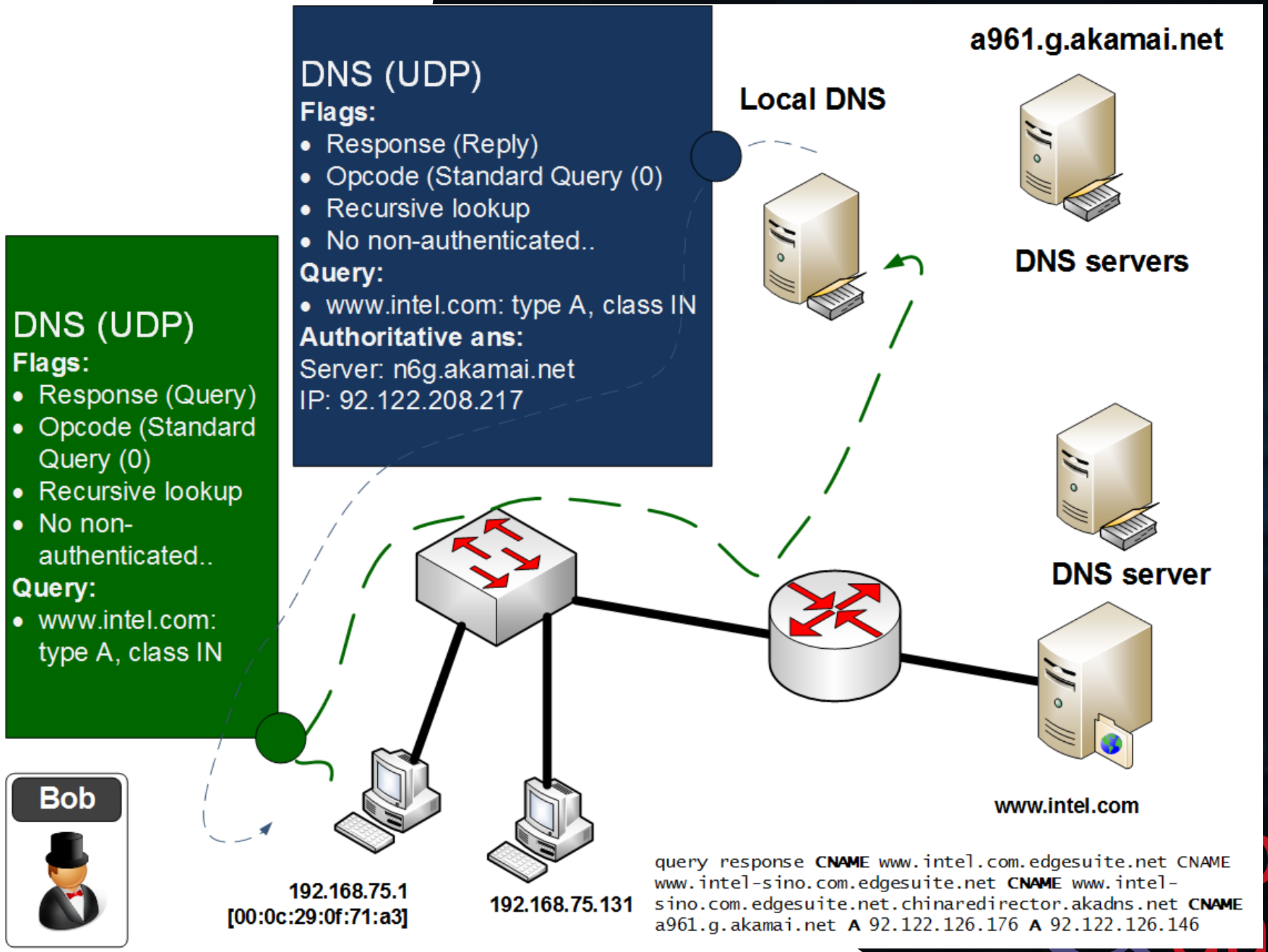
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# DNS

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# DNS


DNS (UDP)
Flags:
• Response (Reply)
• Opcode (Standard Query (0)
• Recursive lookup
• No non-authenticated..
Query:
• www.intel.com: type A, class IN
Authoritative ans:
Server: n6g.akamai.net
IP: 92.122.208.217
DNS (UDP)
Flags:
• Response (Query)
• Opcode (Standard Query (0)
• Recursive lookup
• No non-authenticated..
Query:
• www.intel.com: type A, class IN
a961.g.akamai.net
Local DNS
DNS servers
DNS server
www.intel.com
Bob
192.168.75.1
[00:0c:29:0f:71:a3]
192.168.75.131
query response CNAME www.intel.com.edgesuite.net CNAME www.intel-sino.com.edgesuite.net CNAME www.intel-sino.com.edgesuite.net.chinaredirector.akadns.net CNAME a961.g.akamai.net A 92.122.126.176 A 92.122.126.146

DNS

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# Port Scans

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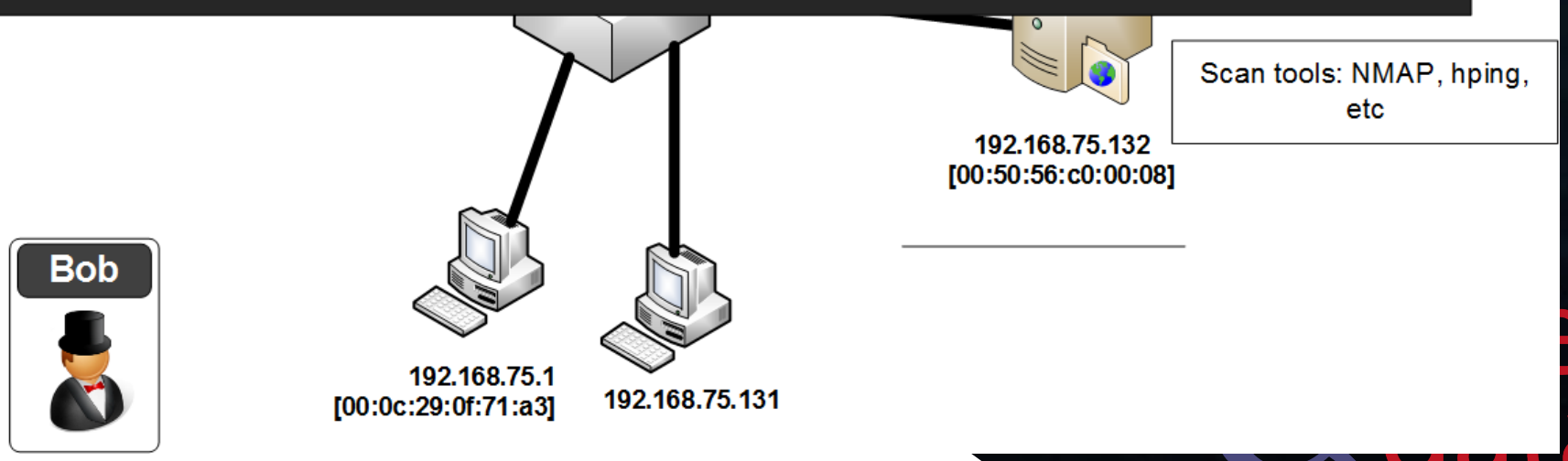
# Port Scans

```
No.     Time        Source                Destination           Protocol Info
     85 25.420710   192.168.75.1          192.168.75.132        TCP      54370 > telnet
[SYN] Seq=0 Win=1024 Len=0 MSS=1460

Frame 85 (58 bytes on wire, 58 bytes captured)
Internet Protocol, Src: 192.168.75.1 (192.168.75.1), Dst: 192.168.75.132
(192.168.75.132)
Transmission Control Protocol, Src Port: 54370 (54370), Dst Port: telnet (23), Seq: 0,
Len: 0

No.     Time        Source                Destination           Protocol Info
     86 25.420836   192.168.75.1          192.168.75.132        TCP      54370 > rap
[SYN] Seq=0 Win=2048 Len=0 MSS=1460

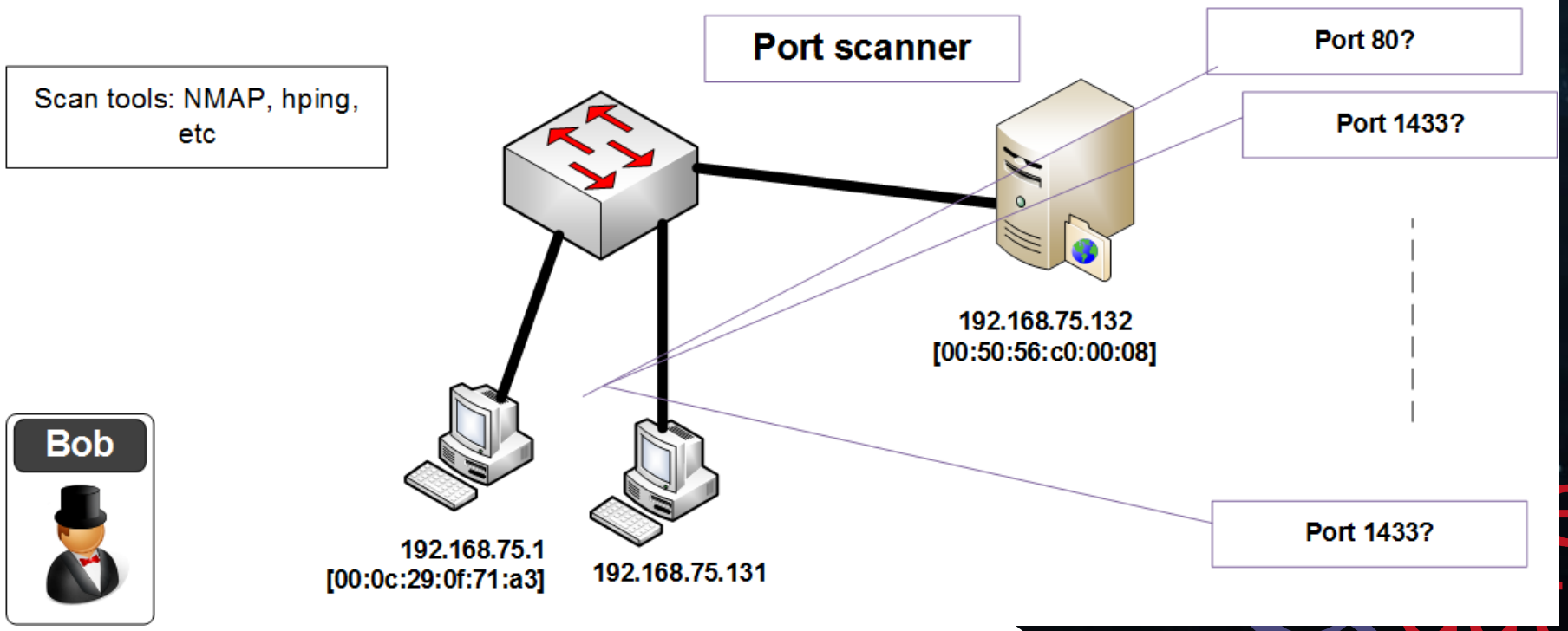
Frame 86 (58 bytes on wire, 58 bytes captured)
Internet Protocol, Src: 192.168.75.1 (192.168.75.1), Dst: 192.168.75.132
(192.168.75.132)
Transmission Control Protocol, Src Port: 54370 (54370), Dst Port: rap (256), Seq: 0,
Len: 0
```



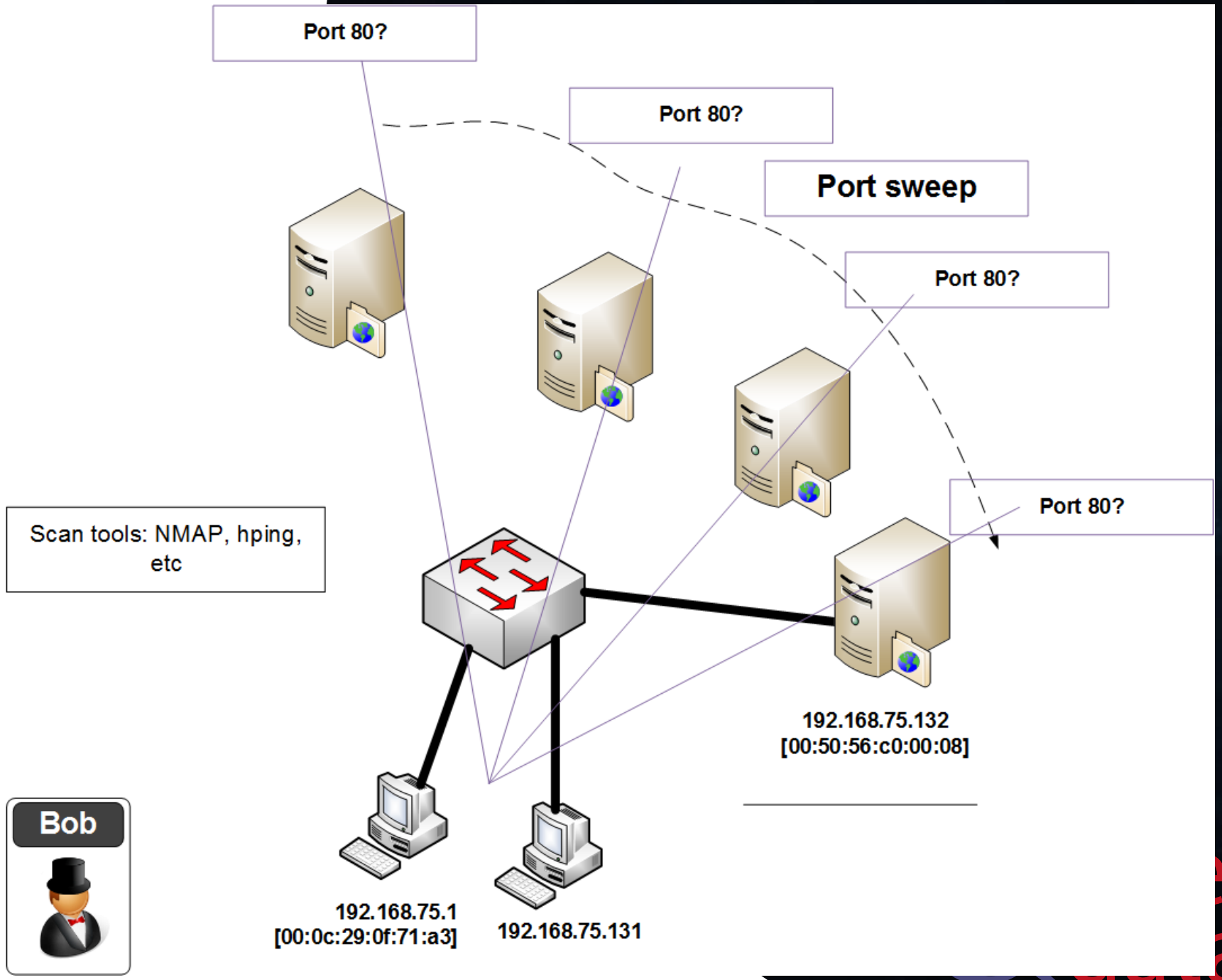

NMAP

# Port Scans

| | | | | |
|---|---|---|---|---|
| TCP Scan | SYN | SYN/ACK | ACK | FIN/ACK |
| SYN Scan | SYN | Creates half-open connections | | |
| FIN Scan | FIN | Closed ports respond with a RST, but open ports ignore. | | |



# Port Scans


Port 80?
Port 80?
Port sweep
Port 80?
Port 80?
Scan tools: NMAP, hping, etc
Bob
192.168.75.132
[00:50:56:c0:00:08]
192.168.75.1
[00:0c:29:0f:71:a3]
192.168.75.131

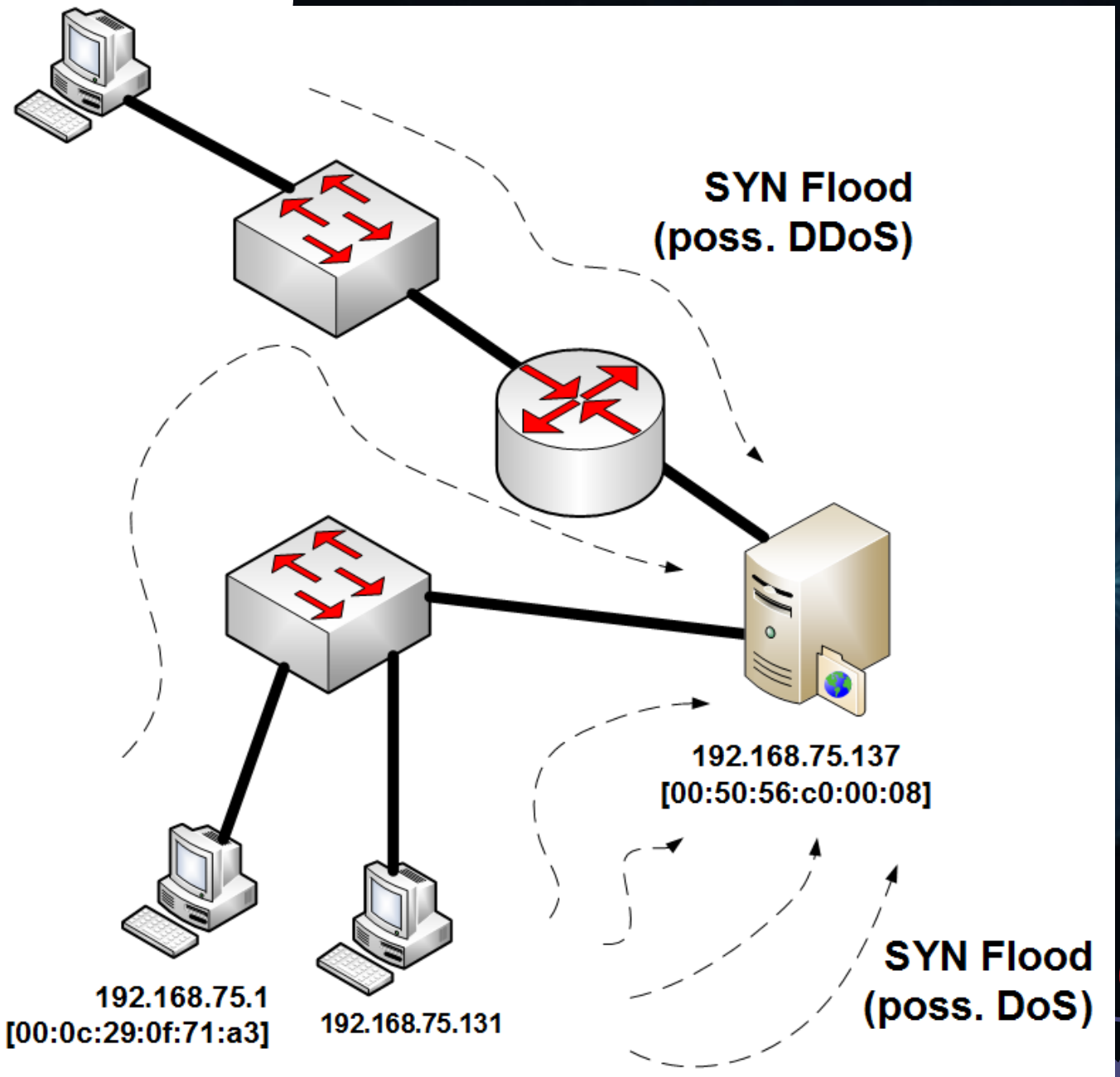
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# SYN Flood

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# SYN Flood




SYN



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# Application Layer

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# Application Layer (HTTP)

```
GET / HTTP/1.1
User-Agent: Opera/9.80 (Windows NT 6.0; U; en) Presto/
2.2.15 Version/10.01
Host: 192.168.75.132
Accept: text/html, application/xml;q=0.9, application/
xhtml+xml, image/png, image/jpeg, image/gif, image/x-
xbitmap, */*;q=0.1 Accept-Language: en-GB,en;q=0.9
Accept-Charset: iso-8859-1, utf-8, utf-16, *;q=0.1
Accept-Encoding: deflate, gzip, x-gzip, identity, *;q=0
Connection: Keep-Alive…
```

```
HTTP/1.1 200 OK
Content-Length: 2606
Content-Type: text/html
Content-Location: http://192.168.75.132/iisstart.htm
Last-Modified: Sun, 13 Dec 2009 15:16:14 GMT
Accept-Ranges: bytes ETag: "fc31243677cca1:745" Server: Microsoft-IIS/
6.0 X-Powered-By: ASP.NET
Date: Sat, 02 Jan 2010 22:33:01 GMT
<HTML> <HEAD> <TITLE>SFC  (Final Test)</TITLE> <META http-equiv=Content-
Type content="text/html; charset=iso-8859-1"> <LINK href="2.css"
type=text/css rel=stylesheet> <style type="text/css"> …
```

HTTP




192.168.75.1
[00:0c:29:0f:71:a3]

192.168.75.131

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# Filters and Regular Expressions

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# Filters

- PNG Filter: http contains "\x89\x50\x4E\x47". Trace with a PNG and PNG filter: Test. Pcap.
- PDF Filter: http contains "%PDF". Trace with a PDF and PDF filter: Test. Pcap
- GIF Filter: http contains "GIF89a". Trace with a GIF and GIF filter: Test. Pcap.
- ZIP Filter: http contains "\x50\x4B\x03\x04". Trace with a ZIP and ZIP filter: Test. Pcap
- JPEG Filter: http contains "\xff\xd8". Trace with a JPEG and JPEG filter: Test. Pcap
- MP3 Filter: http contains "\x49\x44\x33". Trace with an MP3 and MP3 filter: Test. Pcap
- RAR Filter: http contains "\x52\x61\x72\x21\x1A\x07\x00". Trace with a RAR and RAR filter: Test. Pcap
- AVI Filter: http contains "\x52\x49\x46\x46". Trace with a AVI and AVI filter: Test. Pcap
- SWF Filter: http contains "\x46\x57\x53". Trace with a SWF and SWF filter: Test. Pcap
- GZip Filter: http contains "\x1F\x8B\x08". Trace with a GZIP and GZIP filter: Test. Pcap
- Email address Filter: smtp matches ""[a-zA-Z0-9.%+-]+@[a-zA-Z0-9.%+-]"". Trace with an email and Email regex filter: Test. Pcap
- IP address Filter: http matches ""[0-9]{1,3}.[0-9]{1,3}.[0-9]{1,3}.[0-9]{1,3}"". Trace with HTTP traffic and IP address regex filter: Test. Pcap
- Credit card details (Mastercard) Filter: smtp matches ""5\d{3}(\s|-)?\d{4}(\s|-)?\d{4}(\s|-)?\d{4}"". Trace with an email and Mastercard regex filter: Test. Pcap
- Credit card details (Visa) Filter: smtp matches ""4\d{3}(\s|-)?\d{4}(\s|-)?\d{4}(\s|-)?\d{4}"". Trace with an email and Visa filter regex filter: Test. Pcap
- Credit card details (Am Ex) Filter: smtp matches ""3\d{3}(\s|-)?\d{6}(\s|-)?\d{5}"". Trace with an email and Am Ex regex filter: Test. Pcap
- Domain name Filter: http matches ""[a-zA-Z0-9-.]+.(com|org|net|mil|edu|COM|ORG|NET|MIL|EDU|UK)"". Trace with an email and Email regex filter: Test. Pcap
- FTP User/Password Crack Filter: ftp contains "530 User". Trace with FTP Hydra and 530 filter: Test. Pcap
- FTP Login Filter: tcp.port==21 && tcp.flags.syn==1 && tcp.flags.ack==1. Trace with FTP Hydra and SYN/Port 21 filter: Test. Pcap

Filters

# Filters

- tcp.flags.syn==1
- tcp.flags.syn==1 && tcp.flags.ack==0
- http.request.method=="GET”
- udp.port==53



Filters

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